

# Mind the Gap: Improving Transitional Support Materials for Incoming International Mathematics Undergraduates

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July 2026

Durham University Department of Mathematical Sciences Summer Undergraduate Research Award

## 1. Introduction

As our global community develops in technological advancement, undergraduate Mathematics still carries one of the lowest continuation rates in the UK. [4] The National Audit Office identifies one of the most common reasons for dropouts as a lack of prepared understanding of the course, causing demotivation and dissatisfaction. [11] Students are often caught off-guard by university mathematics' higher emphasis on abstract thinking and proof, affecting their view of Mathematics and their own success. [3]

This transition proves especially challenging for international students from high school curricula with different learning styles and norms around mathematical approach. However, international students in first-year mathematics remain relatively invisible in mathematical education research, despite their economic and cultural importance. Thus, it is crucially important to research how to improve the transition into higher education for this large demographic of students, opening a space for international voices in the field of mathematical pedagogy research. By gaining a better understanding of the sociocultural and academic impacts towards international students' transition, educators can strive to improve their overall quality of education more effectively. [9]

Research shows that earlier exposure to university-style mathematical thinking can improve students' transition, influencing their entire university education. [4] At Durham University, the Department of Mathematical Sciences provides students with various pre-university transition materials, including a Blackboard Ultra Induction course to introduce the more social side of the degree, a summer workbook to review prerequisite content, and a diagnostic test to assess the incoming cohort's mathematical abilities. Maintained by Dr Clare Wallace, the summer workbook and diagnostic test have evolved over the last few years to better support students in their academic transition into university mathematics.

However, there are still significant gaps between global high school curricula and skills expected from students early in Analysis I (renowned as the first year module with the highest emphasis on “proof skills” and the steepest transition curve) that are insufficiently filled by these two pre-university resources. This research project identifies misalignments in skills and expectations between international students and lecturers to enhance our department's pre-university resources and transition support. I have also produced suggestions and samples of improvements to the summer workbook and diagnostic test, implementing my research to improve these resources for future cohorts from 2026-27.

## 2. Background

### 2.1 Proofs and Mathematical Teaching

Proof is considered the centre of mathematical thinking and communication. Still, most students in high school and university learn proofs by rote memorisation, without fully understanding the steps or learning the skills to construct a mathematical argument. [13] Lecturers expect these skills to be developed through practice and in-class demonstration. But by presenting finished proofs and supporting lemmas without further explanation, this encourages students to memorise proofs for high school examinations, and fails to show the real mathematical work and grit that

led to this polished result. This provides a misconception of how mathematics is ideally taught at university, and consequently how true mathematicians work. [13]

## 2.2 Differences in International High School Learning Approaches

This transitional problem especially affects many international students, who may face increased difficulty in transition due to different learning styles in high school curricula worldwide, heavily affecting students' perception of competence towards the subject. [3] The mismatch between the mathematical skills they have developed through high school examinations and the mathematical approach lecturers expect them to have creates an additional obstacle for these students. For example, in a study where students in Italy, Switzerland and the UK were surveyed on their transition into 1st year undergraduate mathematics, it is interesting to note that the Swiss and British students struggled with university proofs more than the Italians. This was largely due to the difference in high school curricula. It was found that in the Swiss high schools, Mathematics was treated as an instrument for science and very few proofs were taught. Meanwhile, more than 85% of exam marks went towards calculations and procedures in the UK's A-Levels. Conversely, in the Italian high schools, students were taught deductive mathematical reasoning and approaching proofs using formal language, definitions and axioms, introducing skills more similar to those required at university mathematics. [3]

## 3. Methodology

My research strives to explore how current pre-university transitional materials can be improved to better prepare international students for the mathematical approach in university mathematics. More specifically:

*How can we improve our summer workbook and diagnostic test to better equip incoming international students with the skills expected in early Analysis I?*

My research project is chronologically structured by 4 project objectives:

### Obj 1: Diagnose the Gaps

Through the MATH Taxonomy, I analysed the skills assessed in examination questions from popular incoming international high school curricula (IB, AP, CBSE, Gaokao and HKDSE), the diagnostic test, and Analysis I assessments from the first half of Michaelmas term. Analysis I is the first-year module at Durham University where mathematical skills that are less taught at the school level (eg, proofs, writing) are most notoriously required. Through this comparative investigation, I identified departmental expectations and assumptions of familiarity that are misaligned with the support and training incoming international students receive.

### Obj 2: Design a Solution

I performed a literature review of current pedagogical research, which acted as the theoretical framework to create clear design principles for improving the summer workbook and diagnostic test. Each evidence-based modification is thoroughly explained in this report.

### Obj 3: Create Samples

Through our university STACK platform, I produced sample resources based on the currently provided summer materials, including a revised chapter on proofs for the summer workbook and proof-based questions for the diagnostic test. The purpose of this objective was to create tangible resources that I then submitted to my department as part of a proposal to improve our two main summer preparatory resources.

### Obj 4: Produce Accessible Outputs

In addition to this report, I have produced a poster to present my findings to a wider audience. I have submitted this poster to my department so that, upon approval and implementation of my research into our transitional support material, it can be on display to further inspire future students to research and continue improving the quality of education for all.

## 4. The Problem

### 4.1 MATH Taxonomy

The Mathematical Assessment Task Hierarchy (MATH Taxonomy) is a modification of Bloom’s Taxonomy of Cognitive Skills. Bloom first proposed different categories of assessment questions in general education [12], but several researchers and educators in mathematics have shown preference for the MATH taxonomy from Smith et al., as it has been modified to categorise the different skills required in complete mathematical tasks at the undergraduate level. [2] This newer taxonomy was created due to concerns about how school mathematics exams prepare students for further mathematics studies, ensuring high school students are introduced to and assessed on the required breadth of skills. For this research project, I decided to also use the MATH taxonomy, as my prime focus is also to identify the gaps in skills required to thrive at high school and undergraduate mathematics. The categories in the MATH taxonomy are explained in the table below: [14]

Table 1: The MATH Taxonomy

| Routine procedures               |                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group A                          | <b>Factual Knowledge and Fact Systems [FKFS]</b><br>Recall.<br><br><b>Comprehension [COMP]</b><br>Understanding.<br><br><b>Routine Use of Procedures [RUOP]</b><br>Familiar procedural tasks.                                                                                                                                               |
| New ways                         |                                                                                                                                                                                                                                                                                                                                             |
| Group B                          | <b>Information Transfer [IT]</b><br>Explaining in other visual/numerical/non-technical forms, reordering argument steps.<br><br><b>Application in New Situations [AINS]</b><br>Problem solving in unseen scenarios.                                                                                                                         |
| Construct mathematical arguments |                                                                                                                                                                                                                                                                                                                                             |
| Group C                          | <b>Justifying and Interpreting [JI]</b><br>Find errors and assumptions, limitations of models, prove a theorem to justify a result.<br><br><b>Implications, Conjectures and Comparisons [ICC]</b><br>Make and justify conjectures from result(s).<br><br><b>Evaluation [EV]</b><br>Judge the value of some information for a given purpose. |

Previous research has shown that while we associate the skills in Group C to the abilities mathematicians and problem solvers need, most undergraduate mathematicians lack these skills the most. The University of Edinburgh conducted a study where they applied this taxonomy to A Level, IB, SQA Advanced Higher and their first-year university mathematics modules; their results confirmed that there is a difference between the skills assessed in university and high school. [8] At Oxford University’s Department of Education, Darlington found that most marks in A-Level examinations are awarded for Group A tasks. [2] Her further study comparing pure Mathematics questions in A-Level and Oxford’s undergraduate Mathematics programme showed that while A-Level’s main goal is to assess students’ procedural skills instead of developing their mathematical thinking, in undergraduate mathematics, Group C skills are assessed most, and students need to be able to solve problems and justify solutions. She suggests school-level mathematics should help develop these Group C skills for a successful transition into university study. [1]

### 4.2 Investigation: Justifications

To apply this area of mathematical education research to international students’ transition into undergraduate mathematics, I analysed exam questions from different international high school curricula, Durham’s pre-university mathematics diagnostic test and Analysis I assessments.

The largest number of international students at Durham University comes from China, Hong Kong SAR, the US and India, and the proportions are similar across the rest of the UK. Every year, Durham University accepts an average of around 2500 students from China, 450 from Hong Kong, 280 from the US and 187 from India, while other countries send significantly fewer students. [15] Hence, I believe it is justified to analyse the following high school mathematics curricula:

- International Baccalaureate (IB): Higher Level Analysis and Approaches (HL AA) - the IB students from the US, India, China and Hong Kong combine to more than half of IB's yearly international cohort [6]; HL AA is the Mathematics module required to study Mathematics at Durham.
- Advanced Placement (AP) Calculus BC: the main American education system; AP Calc BC is the Mathematics prerequisite for our course.
- GaoKao: the final secondary school exam in the mainland Chinese education system.
- Central Board of Secondary Education (CBSE) Class XII: the final exams taken in India's main education system.
- Hong Kong Diploma of Secondary Education (HKDSE): the final secondary school exam in Hong Kong.

In this investigation, I took one set of past papers from each curriculum to cover all exam questions in a set of exams. It was more difficult to find past papers for some curricula, so I was not able to keep the exam year consistent across curricula. To cover all exam components, I had to combine exam components from different years for AP Calculus due to limited public resources.

Although this research project aims to improve both the summer workbook and the diagnostic test through the lens of proofs, I only applied the MATH taxonomy to the questions in the diagnostic test. Since there were no questions on proofs in the summer workbook and only one in the diagnostic test, it seemed the most suitable approach to fully capture the proof skills our pre-university transition materials emphasise for students.

As for the choice of undergraduate focus, Analysis I was the most suitable module choice for this investigation because the module places greater emphasis on proofs and is generally regarded as one of the most difficult modules to transition into in Year 1. To assess what proof skills lecturers expect students to have, I decided to analyse all the questions from the assessments in the first half of Michaelmas term (E-Assessments 2 and 4, Gradescope assessments 3 and 5), as the early timing of these assessments imply that lecturers expect such skills and competence from students from the start.

### 4.3 Investigation: Findings

By categorising each exam question using the MATH taxonomy, most high schools do appear to test students mostly on procedural recall. On average, all the high school curricula tested students most on Group A skills (70%), with the highest average of 57% of marks towards Routine Use of Procedures (RUOP). Specifically, HKDSE assigns the highest proportion of marks to RUOP (78%), CBSE allocates the highest 17% to Factual Knowledge and Fact Systems (FKFS), and AP awards the highest 15.6% to Comprehension (COMP). Meanwhile, the only high school curriculum without a significant skew towards Group A is the GaoKao, distributing 34% to Group A, 35% to Group B and 31 to Group C.

Unsurprisingly, the early Michaelmas assessments distribute the highest 55% to JI while still sharing 38% of marks to RUOP. From analysing the questions in Analysis I, it is evident that the key skills most needed and presumed are from Group C (59%), and are (in descending order): Justifying and Interpreting (55%, Group C), Routine Use of Procedures (38%, Group A), Implications, Conjectures and Comparisons (4%, Group C) and Comprehension (4%, Group A).

What is most shocking are the results of the taxonomy analysis of our current diagnostic test, where a striking 87% of marks go to Group A skills (with 57% of total marks to RUOP), 9% to Group B and only 4% (1 JI question) to Group C.

Thus, it is clear from this taxonomy analysis that while most international high schools heavily test on Group A skills, Analysis I at university values Group C skills the most in early assessments. This represents a significant gap between the skills that lecturers expect and the mathematical abilities that students have developed and trained through their high school curricula. This misalignment is mismanaged in our diagnostic test, where we continue to test incoming students in a similar style to high school assessment. It is no wonder that our mathematics students receive a huge shock at the different mathematical approach and assessment style once at university.

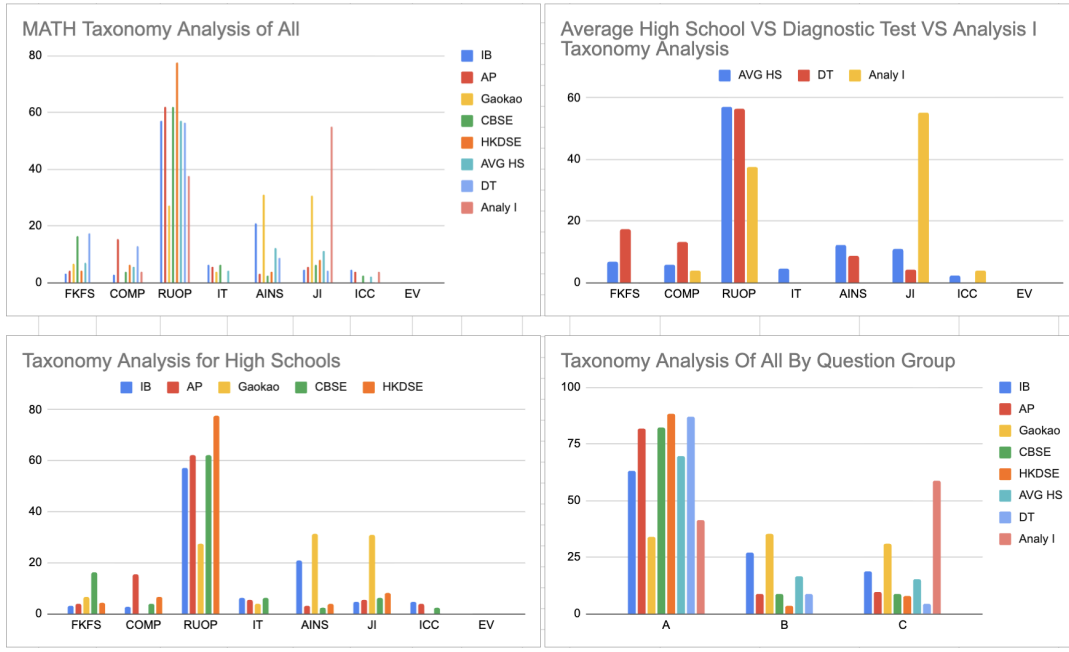


Figure 1: Results of My Comparative Taxonomy Investigation

Data collection and analysis: refer to Figure 1 and [16].

## 5. The Solution

### 5.1 Pedagogy Literature

Current pedagogical literature supports using taxonomy to write questions in order to assess students in a range of skills. Many researchers argue that the types of questions on course assessments show students what skills we value for success in Mathematics and how we expect them to study. Students would adapt their learning approaches to achieve good marks; therefore, it is crucial to use assessments carefully to direct students towards a deep learning approach. [13, 14] In fact, by assessing students only on standard methods and procedures and not on proof, problem solving and exploration, educators contribute to creating a mistaken and distorted impression of Mathematics as a whole. [5] This is especially applicable to our diagnostic test for incoming students. By continuing to use questions similar to high school style to prepare incoming students for university mathematics, universities are reinforcing the misconception that the mathematical approach at university is similar to the procedural and memorisational methods in high school, causing our transitional materials to be ineffective.

The skills required for proofs and the ability to justify results are notoriously hard to teach, delicately balancing the delivery of important factual content with training students to develop their own arguments. So, how can we write multiple-choice questions (MCQs) to further train students in the relevant skills required for Analysis I, specifically justifying and interpreting in proofs and problem solving?

A study from the University of Pretoria suggests that including MCQ's in assessing skill at a higher cognitive level would bring "new dimensions of validity into the assessment of mathematics". It concluded that MCQs are more successful than long response questions in assessing technical, conceptual, modelling and problem-solving skills, while showing no significant advantage in assessing disciplinary recall skills. [5] Mapping these skills onto the MATH taxonomy, these correspond to all the Group A and Group B skills. Although the study showed disfavour towards the skills we would categorise as Group C in the MATH taxonomy, I would note that this was only due to historical data of poorly-written MCQ's and well-written long response questions in this skill area - hence this study does not discourage the use of MCQ's to assess Group C skills; it only suggests that there were not many well-written MCQ's that assess Group C skills in historical data.

Since we have established the importance of Group A and C skills in Analysis I, combined with the evidential lack of preparation of Group C skills in international high schools, it is most reasonable to use our transition materials

to write well-written MCQ's in the topic of proof that specifically target the relevant Group C skills (JI and ICC), increasing the existence of well-written MCQ's that assess Group C skills in historical pedagogy data.

Smith and Petocz suggested a list of subskills educators should assess to encourage students to learn proofs by understanding them and develop problem solving skills. They propose that by using these subskills, students develop the more complex skills needed in constructing a mathematical proof. [13] I mapped their list of subskills onto the MATH taxonomy, and was unsurprised to find that most subskills fall into our relevant Group C skills (JI and ICC). Moreover, by comparing his examples of more complex skills with the questions in Analysis I early Michaelmas assessments, I have found that all early Gradescope assignment questions rely on these complex skills, which are wrongly presumed to be trained from assessing Group C skills in high school. Therefore, Smith and Petocz's suggested subskills act as useful examples for types of practice questions in our summer workbook and diagnostic test.

## 5.2 Design Principles, Suggestions and Explanations

Through my taxonomy analysis and literacy review, I propose the following design principles:

1. Examples should follow departmental expectations of mathematical writing and formatting.
  - a. Thorough explanations should reflect the expected thought process.
  - b. Examples should visually show the expected formatting.
2. Practice questions should train and assess the skills required early in first-year modules.
3. Feedback from practice questions should be thorough, specific and clear.

I have applied these design principles in suggestions for revising the summer workbook and diagnostic test, which I have executed on a copy of the transitional materials using the new departmental Moodle server, ensuring all suggestions are technologically compatible.

Firstly, I have applied Principle 1 to revisions in the Proof Techniques chapter in the summer workbook. Overall, the chapter reminds students of three different kinds of proof techniques (direct, induction and contradiction), giving an example for each. However, I would note that apart from the section for proof by induction, the summer workbook does not cover *when* one would use each type of proof technique. The ability to identify when to use each type of proof technique is very important in university mathematics, where not all questions would suggest the method to use. I also believe there are different aspects of proofs that are worth being mentally prepared for before coming to university, including mathematical writing, structure and use of symbols. So, in the introduction of the chapter, I would clearly state our expectations for incoming students, including understanding when each proof technique is most suitable, justifying steps of a proof clearly using full sentences and structuring working clearly. For each section about each type of proof technique, I would include a few clues to look for in a question that could suggest using the respective method.

In each of the sections about direct proof and proof by contradiction, the example is a large paragraph, but I believe that it is more helpful towards student comprehension to lay the example out in a format similar to an expected answer, with thoroughly explained steps to guide students on the expected thought process. I would also attach a handwritten "model solution" to each example so students can see visually how we would like them to format their answers in assignments and exams.

The example for direct proof also uses an "if and only if" statement without further explanation. As I am aware that many high school curricula do not cover logic, I would include a short explanation of what "if and only if" means in this scenario.

Secondly, I have applied Principle 2 to revising and writing proof-based STACK MCQ's for the summer workbook and diagnostic test. Originally, there were no practice questions on proof in the summer workbook. I believe the summer workbook is the best place to put more practice questions that challenge students to adopt the skills we expect them to have in a formative environment. Hence, I have written three new MCQ questions about proof that utilise the skills presumed in Analysis I:

1. In the first two e-assessments in Analysis I, students are asked to fill in the blanks in a proof by selecting options from a drop-down box. I believe this is helpful to have in the summer workbook to expose students to the style and formatting of university assessment questions. This can also serve as a good COMP (Group A) question, a skill identified as useful in Analysis I. In a similar format, I have written a fill-in-the-blanks question for the proof that  $\sqrt{2}$

is irrational. This question assesses students' abilities to understand the implications of mathematical definitions and follow a proof's line of reasoning.

2. Smith and Petocz suggest testing students in their ability to find mistakes in a proof in their list of subskills required for learning proofs. [13] This would act as a good JI (Group C) question, which is the skill identified as most needed in Analysis I. So, I propose a new MCQ where students must find the first line of a given proof where a mistake first occurs. By finding the mistake, students recognise that when proving a statement, one must start with the given assumptions to conclude and not assume the conclusion to be true at the start, a common mistake amongst high school mathematicians!

3. Smith and Petocz also suggest testing students on using inductive logic to look for results. [13] Although this might not necessarily translate into a proof-based question, I agree this skill is very important, especially to develop an intuition while constructing mathematical arguments. This would also fall under ICC (Group C), another skill identified as expected in early Analysis I assessments. So I have written an MCQ where students have to understand the situation by testing values inductively, recognise a pattern, devise a general formula and apply it to find a result. In my experience, these are all crucial skills required not only in Analysis I but also throughout all first year modules.

In the diagnostic test, there is only one question on proof, which assesses students' ability to identify when it is *not* best to use direct proof. Originally, this assumes students understand when to apply the proof techniques without the summer workbook's coverage. If my suggestions for the proof chapter of the summer workbook are approved, the nature of this question would be fine to leave as is. However, I have realised that the design of the correct MCQ options can cause students to take a wrong approach to this question. When looking at the bank of correct MCQ options in this question, most of these statements would be identified because one would recognise that the statement can be disproven by a counterexample. But since the question asks how to prove the statements and not disprove them, once a student recognises the statement is false, they would automatically know it is the answer. This is not the approach we would like students to take, since this may not require knowledge of identifying when to use proof techniques. Counterexample is also not a method covered in the workbook, thus I would choose not to include it in questions because we tend to prove statements more often than disproving them in Analysis I. So for the correct options in this question, I would stick with the three proof techniques covered in the summer workbook.

Thirdly, I have applied Principle 3 in all new and revised MCQ's, ensuring useful feedback for correct, partially correct and incorrect answers. Much previous research about diagnostic tests shows that feedback from questions is crucial for students to learn from their mistakes and transition better into university. [7, 10] So, in all my proposed questions, I provide thorough feedback with explanations and hints.

Specifically, in the diagnostic test question, I only included one different proof method in the bank of correct MCQ options, so the feedback can be more precise. This is because in STACK, there is no way to tell which random correct answer out of the bank of correct MCQ options is selected for the question. By limiting the list of correct answers to only be proof by induction, I was able to write more useful and specific feedback. Additionally, the system would know which incorrect answer the student has chosen, so I have written feedback specific to the different types of incorrect statements a student could choose.

## 6. Conclusion

Through this research project, I have conducted a thorough taxonomy analysis and identified significant gaps between the skills international students are trained in through high school curricula and the skills lecturers expect them to have at the start of our first-year proof-based module Analysis I. Moreover, the results of my investigation suggest that our pre-university transition materials (summer workbook and diagnostic test) insufficiently address these gaps, encouraging a distorted view of the undergraduate course. Informed by current pedagogical literature regarding teaching proofs, developing problem-solving skills through assessment and writing highly cognitive MCQ's, I have proposed several design principles to improve our transition materials. I then applied these principles by proposing revisions in the summer workbook's chapter on proofs and writing new/revised STACK MCQs that train students in the useful skills for Analysis I identified in my investigation. I utilised our department's Moodle server to actualise these suggestions on duplicate versions of our transitional materials, with thorough explanations and justifications detailed in this report.

I hope this research will provide useful insight into improving our mathematics department's pre-university transitional materials. I also hope to connect with different universities to present these findings and improve the transition into university mathematics in the UK and beyond. I would also like to collaborate with high schools to present

my findings, encourage deeper understanding-based learning in schools and improve school-level mathematics as preparation for higher education. For further study, I believe it would benefit the field of mathematical education to continue this research and broaden this investigation to other aspects of university mathematics and high school curricula. By extension, applying this investigation to different undergraduate mathematics curricula or other subject areas around the world would further encourage improving the transition into higher education as a whole, increasing internationalism in the research field of mathematical education.

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